

Roll No. : _____

CHARTERED INSTITUTE OF TECHNOLOGY, ABU ROAD

B.Tech. I Sem Internal Examination, 2023

INTRODUCTION TO BUILT ENVIRONMENT

1st Year / 1st Sem

1:30 PM TO 03:00 PM

STAFF NAME: - Rajesh Kumar

DATE: 06.11.2023

Maximum Marks: 40

Minimum Marks: 20

Attempt all questions – (each of 2 marks)

10

Q1. What is Infrastructure?

Q2. What is a green building ?

Q3. What is the built environment?

Q4. Write objective of water resources planning.

Q5. What are the uses of steel in building works?

Attempt any two – (each of 8 marks)

16

Q.1 Explain various types of transportation systems.

Q.2 How are masonry works classified? Describe its Different types.

Q.3 Discuss the Elements of BUILT Environment.

Attempt any one – (each of 14 marks)

14

Q1 Write the differences between historical architecture and modern structural design.
Explain building bye-laws and their role in controlled development of built-environment

Q2. What are the general principles in brick masonry? Explain the different bonds used in brick masonry using neat sketches.

CHARTERED INSTITUTE OF TECHNOLOGY, ASU ROAD

Hall No. 1

B.Tech. I Sem IIIrd Internal Examination, 2024

INTRODUCTION TO BUILT ENVIRONMENT

1st Year / 1st Sem

STAFF NAME: - Rajesh Kumar

01:15 PM TO 03:00 PM

DATE: 13-03-2024

Maximum Marks: 70

Attempt all questions - (each of 2 marks)

20

1. What are the factors affecting the built environment?
2. What do you understand about rain water harvesting?
3. Write objectives of water resources planning.
4. What are the components of a smart environment?
5. What are the uses of steel in building works?
6. Define transportation engineering.
7. Differentiate between surveying and leveling.
8. State Hydrological cycle with neat sketch.
9. What is a well-conditioned triangle?
10. State the types of foundation.

Attempt any five - (each of 10 marks)

20

- Q.1 Write the role of transportation with respect to socio-economic conditions?
- Q.2 Explain the components of smart cities.
- Q.3 Explain domestic and home automation with areas of practice of an energy-efficient built environment.
- Q.4 What are the various basic requirements a building should satisfy in design and performance? Explain the main components of a building.
- Q.5 Describe the Internet of Things for the smart built environment.
- Q.6 What do you understand about traffic calming? Write road safety measures for accidents prevention.
- Q.7 Explain physical, chemical and biological characteristics of water as per Indian Standards

Attempt any three - (each of 10 marks)

*Informatory
Dictionary.*

30

- Q.1 What is the modern world approach towards built environment? Explain in detail by giving examples
- Q.2 Explain the types of civil engineering branches with reference to their roles in built environment

Q.3 Write the differences between historical architecture and modern structural design. Explain building bye-laws and their role in controlled development of built-environment?

Q4 What are the components of building physics? Discuss lighting and acoustics aspects of a building

Q.5 Write short notes on any four -

- (a) Smart metering
- (b) Road safety
- (c) Energy flow in ecosystem
- (d) Components of building physics
- (e) Conventional constructions